

- Pascal Vincent, Hugo Larochelle, Yoshua Bengio, and Pierre-Antoine Manzagol. Extracting and composing robust features with denoising autoencoders. In *Proceedings of the 25th international conference on Machine learning*, pp. 1096–1103, 2008.
- Jonas Wallat, Jaspreet Singh, and Avishek Anand. BERTnesia: Investigating the capture and forgetting of knowledge in BERT. *arXiv preprint arXiv:2106.02902*, 2021.
- Xinyou Wang, Zaixiang Zheng, Fei Ye, Dongyu Xue, Shujian Huang, and Quanguan Gu. Diffusion language models are versatile protein learners. *arXiv preprint arXiv:2402.18567*, 2024.
- Joseph L Watson, David Juergens, Nathaniel R Bennett, Brian L Trippe, Jason Yim, Helen E Eisenach, Woody Ahern, Andrew J Borst, Robert J Ragotte, Lukas F Milles, et al. De novo design of protein structure and function with RFDiffusion. *Nature*, 620(7976):1089–1100, 2023.
- Ruibin Xiong, Yunchang Yang, Di He, Kai Zheng, Shuxin Zheng, Chen Xing, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tiejian Liu. On layer normalization in the transformer architecture. In *International Conference on Machine Learning*, pp. 10524–10533. PMLR, 2020.
- Minghao Xu, Zuobai Zhang, Jiarui Lu, Zhaocheng Zhu, Yangtian Zhang, Ma Chang, Runcheng Liu, and Jian Tang. Peer: a comprehensive and multi-task benchmark for protein sequence understanding. *Advances in Neural Information Processing Systems*, 35:35156–35173, 2022.
- Jiasheng Ye, Zaixiang Zheng, Yu Bao, Lihua Qian, and Mingxuan Wang. DINOISER: Diffused Conditional Sequence Learning by Manipulating Noises, 2024. URL <https://arxiv.org/abs/2302.10025>.
- Jason Yim, Brian L Trippe, Valentin De Bortoli, Emile Mathieu, Arnaud Doucet, Regina Barzilay, and Tommi Jaakkola. SE (3) diffusion model with application to protein backbone generation. *arXiv preprint arXiv:2302.02277*, 2023.
- Sheheryar Zaidi, Michael Schaarschmidt, James Martens, Hyunjik Kim, Yee Whye Teh, Alvaro Sanchez-Gonzalez, Peter Battaglia, Razvan Pascanu, and Jonathan Godwin. Pre-training via denoising for molecular property prediction. *arXiv preprint arXiv:2206.00133*, 2022.
- Kaiwen Zheng, Yongxin Chen, Hanzi Mao, Ming-Yu Liu, Jun Zhu, and Qinsheng Zhang. Masked diffusion models are secretly time-agnostic masked models and exploit inaccurate categorical sampling. *arXiv preprint arXiv:2409.02908*, 2024.

A ADDITIONAL MODEL DETAILS

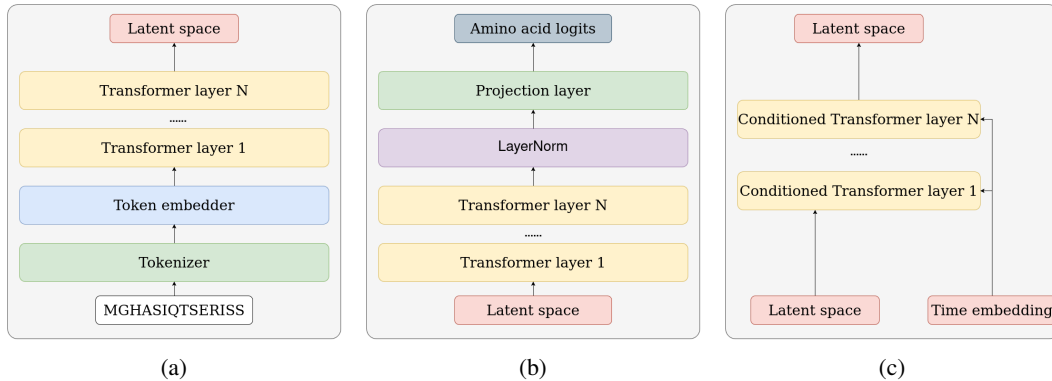


Figure 4: Detailed architecture: (a) encoder, (b) decoder (c) diffusion model.

All three components of the model, encoder-decoder-diffusion, are based on the transformer architecture as illustrated in Fig. 4. We employ RoPE positional encoding Su et al. (2024), Pre-LN Xiong et al. (2020), SwiGLU activation functions Shazeer (2020). Sequences are padded to maximum

sequence length 256, and each component has internal BOS/EOS embeddings which are learned independently and not output. For the decoder we employ a simple projection head onto the amino acid logits. Time-conditioning for the diffusion model is implemented using the adaLN-zero prescription described in DiT Peebles & Xie (2023).

The models are trained on Uniref50 with sequences limited to a maximum length of 254 (+2 for BOS/EOS embeddings), with a batch size of 512, using the AdamW optimizer with weight decay $1e-3$ and learning rate $2e-5$, on one A100 80GB GPU. Model hyper-parameters are provided in Table 3.

Table 3: Model details

Model name	Module	Model size	Channels	Heads	Layers	Steps
MLM-S	Encoder Diffusion	4.7M 7.3M	256	16	6	200k
MLM-M	Encoder Diffusion	18.9M 29.0M	512	16	6	100k
LSD-TN-S	Encoder/Decoder Diffusion	4.7M 7.3M	256	16	6	200k
LSD-TN-M	Encoder/Decoder Diffusion	18.9M 29.0M	512	16	6	100k
LSD-NM-S	Encoder/Decoder Diffusion	4.7M 7.3M	256	16	6	200k
LSD-NM-M	Encoder/Decoder Diffusion	18.9M 29.0M	512	16	6	100k

B EVALUATION DETAILS

In Section 4 we evaluate representation learning on a set of protein property prediction tasks which we adopt from SaProt Su et al. (2023):

- **Thermostability**: protein melting temperature T_m data from the “Human-cell” splits of the Thermostability task of the FLIP benchmark Dallago et al. (2021).
- **HumanPPI**: binary classification whether two proteins interact from HumanPPI data Pan et al. (2010) of the PEER benchmark Xu et al. (2022).
- **Metal Ion Binding**: binary classification of presence of metal ion-binding sites within a protein Hu et al. (2022).
- **DeepLoc**: Predicts subcellular localization of proteins from the DeepLoc dataset Almagro Armenteros et al. (2017).
 - **Subcellular**: multi-class classification identifying one of 10 distinct subcellular compartments.
 - **Binary**: binary classification between membrane-bound or soluble.

We freeze the backbone and train an MLP predictor with a single hidden layer, and either a regressor or classifier head depending on the task. The predictor acts on the mean of the embeddings across the sequence, and for the HumanPPI task we concatenate the mean embeddings of the two proteins. We set the dimension of the predictor’s hidden layer equal to that of the input.

This differs slightly from the SaProt pipeline (also employed by DPLM), which fine-tunes the backbone. As a result the values we obtain for ESM2 are not identical to theirs, nevertheless remain directly comparable.

C ABLATIONS

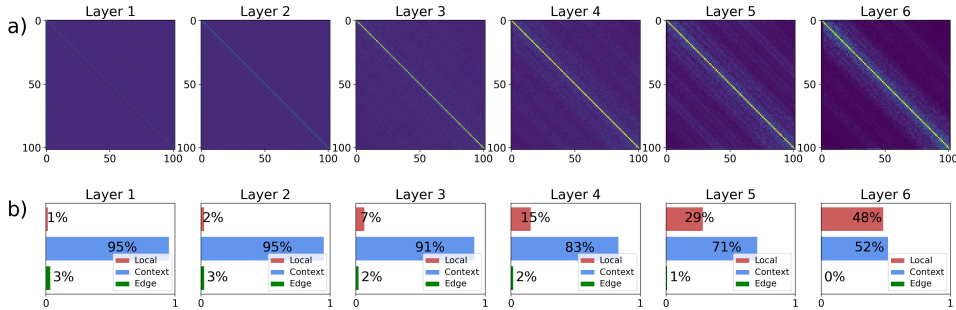
Model	Importance sampling	Modules	Thermostability $\uparrow$	HumanPPI $\uparrow$	Metal Ion Binding $\uparrow$	DeepLoc $\uparrow$	
						Subcellular	Binary
LSD-NM-S	$\checkmark$	Encoder	0.553	62.6	64.1	54.6	77.6
		Diffusion	0.567	60.2	65.0	53.5	76.1
	Off for diffusion	Encoder	0.558	61.7	63.7	54.4	77.3
		Diffusion	0.545	64.7	63.8	52.7	75.8
	Off for noise masking	Encoder	0.543	60.6	62.8	55.2	77.2
		Diffusion	0.540	68.0	59.2	52.2	76.5

Table 4: Importance sampling ablation.

Model	Loss	Modules	Thermostability $\uparrow$	HumanPPI $\uparrow$	Metal Ion Binding $\uparrow$	DeepLoc $\uparrow$	
						Subcellular	Binary
LSD-TN-S	Univariate	Encoder	0.560	58.6	64.6	53.0	76.6
		Diffusion	0.562	62.6	62.8	48.2	75.3
	Multivariate	Encoder	0.548	60.6	63.6	51.6	76.5
		Diffusion	0.528	53.2	61.1	44.9	75.3

Table 5: Normalization loss ablation: we compare the univariate parametric form of the Kullback–Leibler divergence $\frac{1}{2d} \sum_i (\mu_i^2 + \sigma_i^2 - \log \sigma_i^2 - 1)$ to its multivariate counterpart $\frac{1}{2d} (\mu^\top \mu + \Sigma - \log \det \Sigma - d)$, with Σ the covariance matrix.

D CONTEXT LEARNING ANALYSIS

Figure 5: Attention Map Analysis for LSD-NM-S diffusion model. **a)** Average attention logits per layer, aggregated over all heads and 128 protein sequences, each consisting of 100 amino acids. **b)** Distribution of attention scores across different types: *Context* attention, *Local* attention, and edge-token attention.